

the last practice set
chapter 10 calc bc

Test the series for convergence, conditional convergence, or divergence

1. $\sum_{n=2}^{\infty} \frac{2}{n(\ln(n))^3}$

2. $\sum_{n=1}^{\infty} \frac{(2n)^n}{n^{2n}}$

3. $\sum_{n=1}^{\infty} \frac{1}{\ln(n)\ln(n)}$

4. $\sum_{n=1}^{\infty} \frac{(n+5)}{5^n}$

5. $\sum_{n=2}^{\infty} \frac{(-1)^{(n-1)}}{\sqrt{n}-1}$

$$6. \sum_{n=1}^{\infty} (-1)^n \frac{\ln(n)}{\sqrt{n}}$$

$$7. \sum_{n=1}^{\infty} \frac{\cos\left(\frac{n}{2}\right)}{n^2 + 4n}$$

Find radius and interval of convergence

$$8. \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n-1}}{(2n-1)!}$$

$$9. \sum_{n=1}^{\infty} \frac{(-2)^n}{\sqrt{n}} (x + 3)^n$$

Convert these equations to a series

10. $f(x) = \ln(3 + x)$

11. $f(x) = e^{2x}$ in powers of $(x + 1)$

12. Find the Taylor polynomial with remainder, make the approximate, and find the LaGrange Error estimate

$\tan^{-1}(1.1)$, $c=1$, $n=2$